Problem 3: Constant Step Size in SGD

Question. SGD needs $\sum_t \rho_t^2 < \infty$ to converge to the minimum, but in practice people often use a constant step size. Why might this be reasonable in a finite sample setting?

I use a simple example. Take data $y_1, \ldots, y_n$ and the loss $L = \frac{1}{2}(y - \theta)^2$, whose training minimizer is the sample mean $\bar{y}$. With a constant step size ρ, the SGD update is

$$\theta \leftarrow \theta + \rho\,(y_i - \theta).$$

What we actually care about is the risk on new data, $R(\theta) = \mathbb{E}[\frac{1}{2}(Y - \theta)^2]$, which is minimized at the population mean $\mu = \mathbb{E}[Y]$. By the law of large numbers, $\bar{y} \to \mu$ as $n \to \infty$. But with a fixed, finite n, $\bar{y} \neq \mu$ in general. By the CLT, the gap is about

$$\text{statistical error} \;\approx\; \frac{\sigma}{\sqrt{n}}.$$

This error comes only from having n samples. No optimizer can remove it: even if SGD found $\bar{y}$ exactly, we would still be about $\sigma/\sqrt{n}$ away from μ.

So there are two errors: the *statistical error* ($\bar{y}$ vs. μ) and the *optimization error* (θ vs. $\bar{y}$). I want to compare their sizes.

Let $e = \theta - \bar{y}$. Subtracting $\bar{y}$ from both sides of the update gives

$$e_{\text{new}} = (1 - \rho)\,e + \rho\,(y_i - \bar{y}).$$

The first term shrinks the old error. The second term adds new noise every step. Since y_i is a fresh draw, it is independent of e, and $\mathrm{Var}(y_i - \bar{y}) = s^2$ (the variance of the training data). Taking variances:

$$V_{t+1} = (1 - \rho)^2\,V_t + \rho^2 s^2.$$

Because $(1 - \rho)^2 < 1$, V_t settles at a fixed level V where shrinking and adding balance out (like a leaky bucket with water poured in at a constant rate). Solving $V = (1 - \rho)^2 V + \rho^2 s^2$:

$$V = \frac{\rho\, s^2}{2 - \rho} \approx \frac{\rho\, s^2}{2} \quad \Longrightarrow \quad \text{optimization error} \approx s\sqrt{\rho/2}.$$

Note that e itself never stops moving; only the size of its wobble becomes stable.

Since $s \approx \sigma$, the two errors are about the same size when

$$\sqrt{\rho/2} \approx \frac{1}{\sqrt{n}} \quad \Longleftrightarrow \quad \rho \approx \frac{2}{n}.$$

Conclusion

- If $\rho \gg 2/n$, the optimization error dominates, so making ρ smaller helps.

- If ρ is around $2/n$ or smaller, the optimization error is no bigger than the statistical error. Making ρ even smaller lowers the optimization error, but the total error is still controlled by the statistical error, so it barely improves.

- Meanwhile, the cost goes up a lot. The old error shrinks like $(1-\rho)^t$, so to shrink it to a fraction ε we need
$$t \approx \frac{\ln(1/\varepsilon)}{\rho} \qquad (\text{using } \ln(1-\rho) \approx -\rho).$$
The number of steps grows like $1/\rho$. Making ρ 100 times smaller makes the optimization error only 10 times smaller, costs about 100 times more steps, and the total error hardly changes.

So with finite data, a small enough constant step size is "good enough": it gets us within the statistical error we can't avoid anyway, and driving SGD to the exact training minimum would spend a lot of compute for almost no gain on new data.